

# Chapter 4

## One-Electron atoms: Angular Wavefunctions

The angular solution for the hydrogen atom is covered in §2.1.1 of Foot [2].

### 4.1 Angular momentum operator

Returning to the Hamiltonian for a 1-electron atom, we can express the whole thing in spherical coordinates as

$$-\frac{2m}{\hbar^2}H = \frac{1}{r^2} \frac{\partial}{\partial r} \left( r^2 \frac{\partial}{\partial r} \right) + \frac{1}{r^2 \sin(\theta)} \frac{\partial}{\partial \theta} \left( \sin(\theta) \frac{\partial}{\partial \theta} \right) + \frac{1}{r^2 \sin^2(\theta)} \frac{\partial^2}{\partial \phi^2} + Z \frac{2me^2}{4\pi\epsilon_0 r \hbar^2}. \quad (4.1)$$

We will see that this can be simplified using the angular momentum operator,  $\hat{\mathbf{l}}$ , as follows. First, recalling from classical mechanics that

$$\mathbf{l} = \mathbf{r} \times \mathbf{p} \quad (4.2)$$

and from quantum mechanics that translational momentum can be represented in the position representation as  $\hat{\mathbf{p}} \doteq -i\hbar \nabla$ , we can write the Carte-

sian coordinates of the operator  $\hat{\ell} = \hat{\mathbf{r}} \times \hat{\mathbf{p}} \doteq -i\hbar \mathbf{r} \times \nabla$  as

$$\begin{aligned}\hat{\ell}_x &= \hat{y}\hat{p}_z - \hat{z}\hat{p}_y \doteq -i\hbar \left( y \frac{\partial}{\partial z} - z \frac{\partial}{\partial y} \right) \\ \hat{\ell}_y &= \hat{z}\hat{p}_x - \hat{x}\hat{p}_z \doteq -i\hbar \left( z \frac{\partial}{\partial x} - x \frac{\partial}{\partial z} \right) \\ \hat{\ell}_z &= \hat{x}\hat{p}_y - \hat{y}\hat{p}_x \doteq -i\hbar \left( x \frac{\partial}{\partial y} - y \frac{\partial}{\partial x} \right).\end{aligned}\tag{4.3}$$

You can check for yourself to verify that these components satisfy the canonical commutation relations

$$[\hat{\ell}_i, \hat{\ell}_j] = i\hbar \epsilon_{ijk} \hat{\ell}_k \tag{4.4}$$

where  $\epsilon_{ijk}$  is the Levi-Civita<sup>1</sup> symbol (sometimes called the *totally antisymmetric tensor*), which we can take to be defined by

$$\epsilon_{ijk} = \begin{cases} 1 & \text{if } (i, j, k) = (x, y, z), (z, x, y), \text{ or } (y, z, x) \\ -1 & \text{if } (i, j, k) = (x, z, y), (y, x, z), \text{ or } (z, y, x) \\ 0 & \text{for any repeated indices} \end{cases} \tag{4.5}$$

This can be demystified somewhat by noting that the cross product of two vectors in 3D has Cartesian components given by

$$(\mathbf{a} \times \mathbf{b})_i = \sum_{j,k \in \{x,y,z\}} \epsilon_{ijk} a_j b_k \hat{\mathbf{e}}_i.$$

Taking this as the *definition* of the cross product (even if the components of the two vectors don't commute with one another) allows us to write the commutation relations (4.4) in the compact form

$$\hat{\ell} \times \hat{\ell} = i\hbar \hat{\ell}.$$

In spherical coordinates (centered at the center of mass), a bit of calculus can be used to show that

$$\hat{\ell}^2 \doteq -\hbar^2 \left[ \frac{1}{\sin^2(\theta)} \frac{\partial^2}{\partial \phi^2} + \frac{1}{\sin(\theta)} \frac{\partial}{\partial \theta} \left( \sin(\theta) \frac{\partial}{\partial \theta} \right) \right] \tag{4.6}$$

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<sup>1</sup>I *believe* this is pronounced “Leh-vee Chee-vee-tuh.”

so that the Hamiltonian can be written in the form

$$H = -\frac{\hbar^2}{2mr^2} \frac{\partial}{\partial r} \left( r^2 \frac{\partial}{\partial r} \right) + \frac{1}{2m r^2} \hat{\ell}^2 - Z \frac{e^2}{4\pi\epsilon_0 r}. \quad (4.7)$$

Next, we assume that the wavefunctions are separable, which is to say they are products of a purely-radial function,  $R(r)$ , and a purely-angular function,  $A(\theta, \phi)$ , where we solved for the  $s$ -state radial wavefunctions in the previous chapter. Putting  $\psi(r, \theta, \phi) \equiv R(r)A(\theta, \phi)$  into the time-independent Schrödinger equation gives us

$$\begin{aligned} -A \frac{\hbar^2}{2mr^2} \frac{\partial}{\partial r} \left( r^2 \frac{\partial}{\partial r} \right) R + R \frac{1}{2m r^2} \hat{\ell}^2 A - AZ \frac{e^2}{4\pi\epsilon_0 r} R &= E R A \\ A \frac{\partial}{\partial r} \left( r^2 \frac{\partial}{\partial r} \right) R + AZ \frac{2mre^2}{4\pi\epsilon_0 \hbar^2} R + A \frac{2mEr^2}{\hbar^2} R &= R \frac{1}{\hbar^2} \hat{\ell}^2 A. \end{aligned} \quad (4.8)$$

Since the terms on the left side of the equal sign, which depend upon the  $r$ -dependence of  $R(r)$ , must be equal to the term on the right side of the equal sign, which depends on the  $\theta$  and  $\phi$  dependence of  $A(\theta, \phi)$  for all positions  $\mathbf{r}$ , we see that they can each be equal to the same constant (that we will follow Foot in calling  $b$ ) times  $\psi(r, \theta, \phi)$ . This allows us to split them up into two equations, the angular of which is

$$\hat{\ell}^2 A(\theta, \phi) = \hbar^2 b A(\theta, \phi), \quad (4.9)$$

and we see that we seek eigenfunctions  $A(\theta, \phi)$  of  $\hat{\ell}^2$ . Noting further that  $\hat{\ell}_z$  in polar coordinates takes the form

$$\hat{\ell}_z \doteq -i\hbar \frac{\partial}{\partial \phi}, \quad (4.10)$$

we see that

$$[\hat{\ell}^2, \hat{\ell}_z] = 0 \quad (4.11)$$

and these functions must also be eigenfunctions of  $\hat{\ell}_z$ . (It is also true that  $[\hat{\ell}^2, \hat{\ell}_x] = [\hat{\ell}^2, \hat{\ell}_y] = 0$ , but Eq. (4.4) shows that  $A$  can only be an eigenfunction of one of the Cartesian components at a time, and so we choose  $\ell_z$  by convention.)

Before moving on to finding the eigenvalues and eigenstates of  $\hat{\ell}^2$  and  $\hat{\ell}_z$ , I am going to state without proof that the same commutation relations (4.4) apply to spin, and so the algebra we are encountering will apply to both spin and orbital angular momentum. We will use  $\hat{\mathbf{j}}$  ( $\hat{\mathbf{J}}$ ) to represent a dimensionful (dimensionless) general angular momentum and  $\hat{\ell}$  ( $\hat{\mathbf{L}}$ ) specifically for dimensionful (dimensionless) orbital angular momentum. Since we are not yet including spin,  $\hat{\mathbf{L}} = \hat{\mathbf{J}}$  and we will use both more-or-less interchangeably until we introduce spin.<sup>2</sup>

In terms of the dimensionless operators, we can write (4.9) as

$$\hat{\mathbf{L}}^2 A(\theta, \phi) = b A(\theta, \phi), \quad (4.12)$$

which is the position representation of the more abstract eigenvector statement

$$\hat{\mathbf{L}}^2 |\psi\rangle = b |\psi\rangle.$$

## 4.2 Angular momentum eigenvalues

At this point, we're going to develop some useful algebra for angular momentum to solve for the properties of these eigenvectors. Since this material will generalize to other types of angular momenta, I will switch to using  $\hat{\mathbf{J}}$  instead of  $\hat{\mathbf{L}}$  for this section so that we can apply the results later.

We are tasked with finding the eigenvalues and simultaneous eigenvectors of  $\hat{\mathbf{J}}^2$  and  $\hat{J}_z$ , which we might as well denote  $|JM\rangle$  with some yet-to-be-determined labels  $J$  and  $M$  such that

$$\hat{\mathbf{J}}^2 |JM\rangle = \lambda(J) |JM\rangle \quad (4.13)$$

$$\hat{J}_z |JM\rangle = M |JM\rangle \quad (4.14)$$

where  $\lambda(J)$  and  $M$  are the yet-to-be-determined eigenvalues of  $\hat{\mathbf{J}}^2$  and  $\hat{J}_z$ . Here,  $J$  (just like  $M$ ) is just a number, whereas  $\hat{J}_z$  and  $\hat{\mathbf{J}}$  are operators.  $J$  is something like an eigenvalue, but is more precisely just a label; the actual eigenvalue of  $\hat{\mathbf{J}}^2$  is some function of  $J$  that we're calling  $\lambda(J)$ .

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<sup>2</sup>Specifically, I will tend to use  $\hat{\mathbf{L}}$  when working with spatial wavefunctions and  $\hat{\mathbf{J}}$  when working with abstract statements whose results may generalize beyond just orbital angular momentum.

It will be helpful for us to define the following linear combinations of Cartesian components,

$$\hat{J}_+ \equiv \hat{J}_x + i\hat{J}_y \quad (4.15)$$

$$\hat{J}_- \equiv \hat{J}_x - i\hat{J}_y. \quad (4.16)$$

Note that these are not Hermitian;  $\hat{J}_+^\dagger = \hat{J}_-$ .

What do these do? Well, we can first use (4.4) to show that

$$\begin{aligned} [\hat{\mathbf{J}}^2, \hat{J}_\pm] &= 0 \\ [\hat{J}_z, \hat{J}_\pm] &= \pm \hat{J}_\pm \\ [\hat{J}_+, \hat{J}_-] &= 2\hat{J}_z. \end{aligned} \quad (4.17)$$

We can now find out what the eigenvalues of the kets  $(\hat{J}_\pm |J M\rangle)$  are:

$$\begin{aligned} \hat{\mathbf{J}}^2 (\hat{J}_\pm |J M\rangle) &= \hat{J}_\pm \hat{\mathbf{J}}^2 |J M\rangle = \lambda(J) (\hat{J}_\pm |J M\rangle) \\ \hat{J}_z (\hat{J}_\pm |J M\rangle) &= (\hat{J}_\pm \hat{J}_z \pm \hat{J}_\pm) |J M\rangle = (M \pm 1) (\hat{J}_\pm |J M\rangle). \end{aligned} \quad (4.18)$$

So we see that  $\hat{J}_\pm |J M\rangle$  are (potentially unnormalized) eigenkets of  $\hat{\mathbf{J}}^2$  and  $\hat{J}_z$  with eigenvalues  $\lambda(J)$  and  $(M \pm 1)$ . So  $\hat{J}_+$  “raises” the  $\hat{J}_z$  eigenvalue  $M$  by 1 and  $\hat{J}_-$  “lowers” it by 1. For this reason, these are sometimes referred to as the raising and lowering operators for the  $\hat{\mathbf{e}}_z$ -projection of the angular momentum.

Let’s look more closely at their normalization:

$$\begin{aligned} \langle J M | \hat{J}_- \hat{J}_+ |J M\rangle &= \langle J M | (\hat{J}_x - i\hat{J}_y)(\hat{J}_x + i\hat{J}_y) |J M\rangle \\ &= \langle J M | (\hat{J}_x^2 + \hat{J}_y^2 - i\hat{J}_y\hat{J}_x + i\hat{J}_x\hat{J}_y) |J M\rangle \\ &= \langle J M | (\hat{J}_x^2 + \hat{J}_y^2 + i[\hat{J}_x, \hat{J}_y]) |J M\rangle \\ &= \langle J M | (\hat{J}_x^2 + \hat{J}_y^2 - \hat{J}_z) |J M\rangle \\ &= \langle J M | (\hat{\mathbf{J}}^2 - \hat{J}_z^2 - \hat{J}_z) |J M\rangle \\ &= \langle J M | (\hat{\mathbf{J}}^2 - \hat{J}_z(\hat{J}_z + 1)) |J M\rangle \\ &= \lambda(J) - M(M + 1). \end{aligned} \quad (4.19)$$

Even though it may not be normalized, the kets  $(\hat{J}_\pm |J M\rangle)$  are vectors in Hilbert space and their norms have to be non-negative. We therefore conclude

that there must be some maximum allowed value of  $M = M_{\max}$  such that  $|J M_{\max}\rangle$  can no longer be raised by  $\hat{J}_+$ :

$$\hat{J}_+ |J M_{\max}\rangle = 0. \quad (4.20)$$

This state is called the *M-stretched state*. Applying Eq. (4.19) to  $|J M_{\max}\rangle$  tells us that

$$\lambda(J) = M_{\max}(M_{\max} + 1). \quad (4.21)$$

A similar procedure can be used to conclude that there is a minimum  $M = M_{\min}$  and

$$\lambda(J) = M_{\min}(M_{\min} - 1). \quad (4.22)$$

To determine what these are, the standard procedure is to start counting the number of different allowed values of  $M$  from  $M_{\min}$  to  $M_{\max}$ . About a page of somewhat tedious algebra can be used to conclude that

$$\hat{\mathbf{J}}^2 |J M\rangle = J(J + 1) |J M\rangle \quad (4.23)$$

$$\hat{J}_+ |J M\rangle = \sqrt{J(J + 1) - M(M + 1)} |J, M + 1\rangle \quad (4.24)$$

$$\hat{J}_- |J M\rangle = \sqrt{J(J + 1) - M(M - 1)} |J, M - 1\rangle \quad (4.25)$$

with  $M$  taking values from  $M_{\min} = -J$  to  $M_{\max} = J$  is a workable solution.<sup>3</sup> We now have a set of *angular momentum eigenstates*  $\{|J M\rangle\}$  and their associated eigenvalues for operators  $\hat{\mathbf{J}}^2$  and  $\hat{J}_z$ .

Note that  $\hat{\mathbf{J}}^2 |J M\rangle = J(J + 1) |J M\rangle$  also tells us that the constant  $b$  in Eq. (4.12), is  $b = L(L + 1)$  for eigenstates of  $\hat{\mathbf{L}}^2$  and  $\hat{L}_z$ , allowing us to write the Schrödinger equation in the form

$$\begin{aligned} A_{LM} \frac{\partial}{\partial r} \left( r^2 \frac{\partial}{\partial r} \right) R + A_{LM} Z \frac{2mre^2}{4\pi\epsilon_0\hbar^2} R + A_{LM} \frac{2mEr^2}{\hbar^2} R &= R A_{LM} L(L + 1) \\ \frac{d}{dr} \left( r^2 \frac{d}{dr} \right) R_L(r) + Z \frac{2mre^2}{4\pi\epsilon_0\hbar^2} R_L(r) + \frac{2mEr^2}{\hbar^2} R_L(r) &= R_L(r) L(L + 1) \end{aligned} \quad (4.26)$$

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<sup>3</sup>There is a phase convention that was chosen here that ensures that the eigenvalues of  $\hat{J}_x$  are real and the eigenvalues of  $\hat{J}_y$  are imaginary.

### 4.3 Angular momentum eigenstates

Having constructed a complete set of orthonormal angular momentum eigenstates

$$\langle J' M' | J M \rangle = \delta_{J'J} \delta_{M'M}, \quad (4.27)$$

we seek the properties of these states. Once again, we are going to use a minimalist approach to combine some intuition with a couple of results that we will not prove to argue our way to the correct solution without actually going through the differential equations rigmarole.

For orbital angular momentum, Eq. (4.10) combined with (4.14) suggests that

$$A_{LM}(\boldsymbol{\omega}) \propto e^{iM\phi} \quad (4.28)$$

where periodic boundary conditions dictate that  $M$  (and therefore  $L$ ) takes integer values. As before  $\boldsymbol{\omega}$  denotes both  $\theta$  and  $\phi$ , and is equivalent to  $\hat{\mathbf{e}}_r$  – these are interchangeably used to indicate coordinates on the surface of the unit sphere. For us, though they are completely interchangeable and mean the same thing,  $\boldsymbol{\omega}$  is perhaps more suggestive of spherical coordinates and  $\hat{\mathbf{e}}_r$  may be more suggestive of Cartesian coordinates.

For the  $\theta$  dependence, we again use some intuition to suggest that the ground state will not have any nodes in  $\theta$  since nodes imply wiggles (and more wiggles imply that the spatial derivative, which is momentum, is larger, and so therefore is the energy). The ground state,  $|0,0\rangle$  in the  $|L,M\rangle$  basis, could have a spatial wavefunction that is just a constant:  $A_{0,0}(\boldsymbol{\omega}) = \text{const.}$ . Normalization requires that the integral over solid angle is 1,

$$\int d^2\boldsymbol{\omega} A^*(\boldsymbol{\omega}) A(\boldsymbol{\omega}) = \int_0^{2\pi} d\phi \int_0^\pi d\theta \sin(\theta) A^*(\theta, \phi) A(\theta, \phi) \quad (4.29)$$

and we find that

$$A_{0,0}(\boldsymbol{\omega}) = \sqrt{\frac{1}{4\pi}} \quad (4.30)$$

is a solution.

For  $L = 1$ , you likely suspect that we will be adding a single node, which means that we will have  $A_{1,M}$  proportional to  $\sin(\theta)$  or  $\cos(\theta)$  since we require periodic boundary conditions in  $\theta \rightarrow \theta + 2\pi$ . For larger values of  $L$ , we can expect  $L$  nodes in the  $\theta$  dependence. Since the  $M$ -stretched state's spatial

wavefunction for a state with orbital angular momentum quantum number  $L$  is going to be proportional to  $e^{iL\phi}$  and we are likely to have  $L$  nodes in the  $\theta$  dependence, this suggests two possibilities as the first to try for wavefunctions of the stretched state,  $\sin^L(\theta)e^{iL\phi}$  and  $\cos^L(\theta)e^{iL\phi}$ . To figure out if either of these will work, we can look at the spatial representation of the orbital angular momentum projection-raising operator,

$$\hat{L}_+ \doteq e^{i\phi} \left( \frac{\partial}{\partial \theta} + i \cot(\theta) \frac{\partial}{\partial \phi} \right) \quad (4.31)$$

and we find that

$$\begin{aligned} \hat{L}_+ \sin^L(\theta) e^{iL\phi} &= e^{i\phi} \left( \frac{\partial}{\partial \theta} + i \cot(\theta) \frac{\partial}{\partial \phi} \right) \sin^L(\theta) e^{iL\phi} \\ &= e^{i\phi} \left( L \cos(\theta) \sin^{L-1}(\theta) + i \cot(\theta) \sin^L(\theta) \frac{\partial}{\partial \phi} \right) e^{iL\phi} \\ &= e^{i(L+1)\phi} L \left( \cos(\theta) \sin^{L-1}(\theta) - \cos(\theta) \sin^{L-1}(\theta) \right) \\ &= 0 \end{aligned} \quad (4.32)$$

whereas

$$\begin{aligned} \hat{L}_+ \cos^L(\theta) e^{iL\phi} &= e^{i\phi} \left( \frac{\partial}{\partial \theta} + i \cot(\theta) \frac{\partial}{\partial \phi} \right) \cos^L(\theta) e^{iL\phi} \\ &= e^{i\phi} \left( -L \sin(\theta) \cos^{L-1}(\theta) + i \cot(\theta) \cos^L(\theta) \frac{\partial}{\partial \phi} \right) e^{iL\phi} \\ &= e^{i(L+1)\phi} L \left( -\sin(\theta) \cos^{L-1}(\theta) - \frac{\cos^{L+1}(\theta)}{\sin(\theta)} \right) \\ &\neq 0. \end{aligned} \quad (4.33)$$

And so we have found that  $A_{L,L}(\boldsymbol{\omega}) \propto \sin^L(\theta)e^{iL\phi}$  indeed works. Normalizing this yields

$$A_{L,L}(\boldsymbol{\omega}) = \frac{(-)^L}{L! 2^L} \sqrt{\frac{(2L+1)!}{4\pi}} \sin^L(\theta) e^{iL\phi}. \quad (4.34)$$

In principle, Eq. (4.34) and Eq. (4.25) are all that are needed to find all of the angular wavefunctions. Repeated application of  $\hat{L}_-$ ,

$$\hat{L}_- \doteq e^{-i\phi} \left( -\frac{\partial}{\partial \theta} + i \cot(\theta) \frac{\partial}{\partial \phi} \right), \quad (4.35)$$



to the stretched wavefunction for  $|L, L\rangle$  will give all of the wavefunctions in the  $\{|LM\rangle\}$  manifold. However, this is cumbersome, and there are algebraic ways to skip straight to the answers, which we will provide without proof.

The upshot is that the angular spatial wavefunctions are special functions called *spherical harmonics*, denoted by  $Y_{LM}(\boldsymbol{\omega})$ . Just to be complete, here, we can mention that these functions are given by

$$Y_{LM}(\boldsymbol{\omega}) = (-)^M \sqrt{\frac{2L+1}{4\pi} \frac{(L-M)!}{(L+M)!}} P_L^M[\cos(\theta)] e^{iM\phi} \quad (4.36)$$

where  $P_b^a[x]$  is an associated Legendre polynomial in  $x$  (without the ‘‘Condon-Shortley phase’’<sup>4</sup>). This definition implies that

$$Y_{LM}^*(\boldsymbol{\omega}) \equiv (Y_{LM}(\boldsymbol{\omega}))^* = (-1)^M Y_{L,-M}(\boldsymbol{\omega}). \quad (4.37)$$

Since the spherical harmonics are wavefunctions for angular momentum eigenstates, we could relate them to state vectors via

$$\begin{aligned} \langle \boldsymbol{\omega} | L M \rangle &= Y_{LM}(\boldsymbol{\omega}) = (-1)^M Y_{L,-M}^*(\boldsymbol{\omega}) \\ \langle L M | \boldsymbol{\omega} \rangle &= (-1)^M Y_{L,-M}(\boldsymbol{\omega}) = Y_{L,M}^*(\boldsymbol{\omega}). \end{aligned}$$

The spherical harmonics are an orthonormal basis in the angular coordinates  $\boldsymbol{\omega} = (\theta, \phi)$ :

$$\int_0^{2\pi} d\phi \int_0^\pi d\theta \sin(\theta) Y_{L'M'}^*(\boldsymbol{\omega}) Y_{LM}(\boldsymbol{\omega}) = \delta_{L'L} \delta_{M'M}. \quad (4.38)$$

They also obey the relationships we require for orbital angular momentum eigenstates,

$$\begin{aligned} \hat{\mathbf{L}}^2 Y_{LM}(\boldsymbol{\omega}) &= L(L+1) Y_{LM}(\boldsymbol{\omega}) \\ \hat{L}_z Y_{LM}(\boldsymbol{\omega}) &= M Y_{LM}(\boldsymbol{\omega}) \\ \hat{L}_\pm Y_{LM}(\boldsymbol{\omega}) &= \sqrt{L(L+1) - M(M \pm 1)} Y_{L,M \pm 1}(\boldsymbol{\omega}). \end{aligned} \quad (4.39)$$

For our purposes, it will often suffice to simply note that these special functions provide a representation of  $|L M\rangle$  and if one ever needs to use explicit forms, they are easily available in tables and as built-in functions of various programming languages.

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<sup>4</sup>The Condon-Shortley phase is a factor of  $(-)^M$  that is sometimes included in the definition of the associated Legendre polynomials. Basically, the Condon-Shortley phase convention *for the wavefunction* results in a factor of  $(-1)^M$  on the wavefunction, so we don’t want an additional factor of  $(-)^M$  from the associated Legendre polynomial if there’s a  $(-)^M$  in the definition of  $Y_{LM}$  in terms of  $P_L^M$ .

## 4.4 Radial Wavefunctions for non- $s$ -state electrons

We can now revisit the radial wavefunction for states with nonzero  $L$  since we know that the eigenvalue of  $\hat{\ell}^2$  is  $\hbar^2 L(L+1)$ . If we consider how this term appears in the Hamiltonian,

$$H = -\frac{\hbar^2}{2mr^2} \frac{\partial}{\partial r} \left( r^2 \frac{\partial}{\partial r} \right) + \frac{\hbar^2 L(L+1)}{2m r^2} - Z \frac{e^2}{4\pi\epsilon_0 r}, \quad (4.40)$$

we see that it acts just like a conservative potential. However, we can also see that it gets *larger* as  $r \rightarrow 0$ , in contrast to the Coulomb potential. This effective potential due to nonzero angular momentum is known as a *centrifugal barrier*. It allows us to take a 3D (or 2D) problem and map it onto an effective 1D problem where we only need to consider the radial coordinate.

The radial ODE, equation (4.26), can be written in the form

$$\begin{aligned} \frac{1}{r^2} \frac{d}{dr} \left( r^2 \frac{d}{dr} \right) R_L(r) + 2 \frac{Z}{a_0} \frac{1}{r} R_L(r) + \frac{2mE}{\hbar^2} R_L(r) &= R_L(r) \frac{L(L+1)}{r^2} \\ \left[ \frac{d^2}{dr^2} + \frac{2}{r} \frac{d}{dr} + 2 \frac{Z}{a_0} \frac{1}{r} - \frac{L(L+1)}{r^2} \right] R_L(r) &= -\frac{2mE}{\hbar^2} R_L(r). \end{aligned} \quad (4.41)$$

Recalling our earlier arguments for  $s$  states that led us to find that the radial wavefunction of the  $ns$  state is given by  $e^{-\alpha r}$  times a polynomial of degree  $n-1$  in  $r$ , we can ask if there will be a solution of this form for  $n=1$  with  $L > 0$ . Since the  $s$  state radial wavefunction for  $n=1$  satisfies the equation above only if  $L=0$ , we conclude that there is no solution with a zeroth-degree polynomial (i.e. a constant) coefficient for  $L \neq 0$ . We interpret this as saying that there are no solutions with  $L \neq 0$  and  $n=1$ .

For  $n=2$ , we allow a first-degree polynomial in  $r$  (times  $e^{-\alpha r}$ ) as before,

$$p_1(r) = \left( -b_0 \frac{a_0}{Z} + b_1 r \right). \quad (4.42)$$

A slight adaptation of our process for  $s$  states gives a differential equation for our polynomial

$$\left[ \frac{d^2}{dr^2} + \left( \frac{2}{r} - 2\alpha \right) \frac{d}{dr} + 2 \left( \frac{Z}{a_0} - \alpha \right) \frac{1}{r} - L(L+1) \frac{1}{r^2} \right] p(r) = 0. \quad (4.43)$$

For  $p_1(r)$ , we find

$$0 = \left(\frac{2}{r} - 2\alpha\right) b_1 + 2 \left(\frac{Z}{a_0} - \alpha\right) b_1 - 2 \left(\frac{Z}{a_0} - \alpha\right) b_0 \frac{a_0}{Z} \frac{1}{r} - L(L+1) b_1 \frac{1}{r} + L(L+1) b_0 \frac{a_0}{Z} \frac{1}{r^2}. \quad (4.44)$$

Since the only term that goes as  $1/r^2$  is the last term, we know that the coefficient has to vanish, from which we deduce that  $b_0 = 0$  (assuming  $L \neq 0$ ). The terms that are independent of  $r$  are the same as they were for the  $2s$  state, and we again have

$$\alpha = \frac{Z}{2a_0} \quad (4.45)$$

and

$$E = -Z^2 \frac{e^2}{8\pi\epsilon_0 a_0} \times \frac{1}{2^2}. \quad (4.46)$$

The remaining nonzero terms that go as  $1/r$  yield

$$0 = \frac{2}{r} b_1 - \frac{L(L+1)}{r} b_1$$

$$L(L+1) = 2 \quad (4.47)$$

and we see that  $L = 1$ , which means we have a  $2p$  state. To get  $b_1$ , we enforce normalization of the radial integral (since  $A_{1,M}(\theta, \phi) \equiv Y_{1,M}(\theta, \phi)$  is normalized for the angular integrals):

$$1 = \int_0^\infty dr r^2 |b_1 r e^{-Zr/2a_0}|^2 = |b_1|^2 \frac{24a_0^5}{Z^5} \quad (4.48)$$

and, choosing a phase, we have

$$b_1 = \sqrt{\frac{Z^5}{24a_0^5}}. \quad (4.49)$$

The complete wavefunctions for the  $n = 2$ ,  $L = 1$  states with projection quantum number  $M$  are therefore

$$\psi_{2,1,M}(\mathbf{r}) = \sqrt{\frac{Z^5}{24a_0^5}} r e^{-Zr/2a_0} Y_{1,M}(\boldsymbol{\omega}). \quad (4.50)$$

While we have not demonstrated this for arbitrary  $n$ , the fact that the energy eigenvalues of the  $2s$  and  $2p$  states are the same is a general trend at this level of theory – the energies are independent of  $L$ . When we add the spin-orbit interaction, we will see that this is no longer the case.

